Vector

it's magnitude.

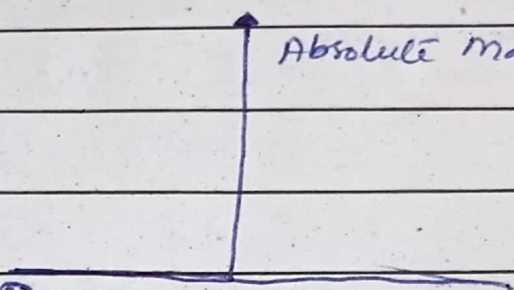
Lec # 12:-

Local Maximum and Minimum:-

Absolute maximum:- 1

highest point of func.

Local Maximum. 2



Local Minimum 3

Absolute minimum 4

Local Maximum and minimum values:-

A fn: of two variables has a local maxima at (a,b). If $\boxed{f(x,y) \leq f(a,b)}$ then $f(a,b)$ is local maximum value.

A fn: of two variables has local minimum at (a,b). If $\boxed{f(x,y) \geq f(a,b)}$ then $f(a,b)$ is local minimum value.

Day: M T W T F S

Date:

Critical Point :-

funct is local maximum or local minimum , and derivative exist. → will be 0 / defined.

Fermat's theorem :- (Multivariable)

If f has a local max : or min : at (a, b) and f_x and f_y exists there ; then $f_x(a, b) = 0$ and $f_y(a, b) = 0$

$\boxed{f_x(a, b) = 0 \text{ or } f_y(a, b) = 0}$ → Critical point or Stationary point

Saddle point :- A point where slop of f is zero or local neither maxima nor minima.

Q. Find critical point of $f(x, y) = x^2 + y^2 - 2x - 6y + 14$

$f(x) = 2x + 0 - 2 - 0 + 0$

$= 2x - 2 = 0$

$= x = \frac{\cancel{2}}{\cancel{2}}$

$\boxed{x = 1}$

$f(y) = 0 + 2y - 0 - 6 + 0$

$= 2y - 6 = 0$

$y = \frac{\overset{3}{\cancel{6}}}{\cancel{2}}$

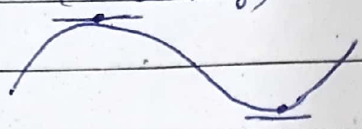
$\boxed{y = 3}$ So, the only critical point is $(1, 3)$

Slope = derivative

Date: ______ Day: M T W T F S

Saddle point (slope = 0) ~~no maximum point~~ no minimum point

(increasing)



(decreasing)

Exp:- find critical points of following func

a) $f(x, y) = 2x^2 + 4y^2 - 4x - 6y + 9.$

Solution:-

$$f(x) = 4x + 0 - 4 - 0 + 0$$

$$= 4x - 4 = 0$$

$$= x = \frac{\cancel{4}}{\cancel{4}}$$

$$\boxed{x = 1}$$

$$f(y) = 0 + 8y - 0 - 6 + 0$$

$$= 8y - 6 = 0$$

$$= y = \frac{\cancel{6}^{3}}{\cancel{8}_{4}}$$

$$\boxed{y = \frac{3}{4}}$$

So the only critical point is $\left(1, \frac{3}{4}\right)$

Scanned with CamScanner

Date:________ Day: M T W T F S

b) $f(x,y) = x^2 + 9y^2 + x - y - 3.$

Solution:-

~~0 + 180~~

$f(x) = 2x + 0 + 1 - 0$

$= 2x + 1$

$\boxed{x = \frac{-1}{2}}$

$f(y) = 0 + 18y + 0 - 1 - 0$

$= 18y - 1$

$= \boxed{y = \frac{1}{18}}$

So, the only critical point is $\left(\frac{-1}{2}, \frac{1}{18}\right)$